VULCAN AUTONOMOUS AI-TO-SILICON SERIES DOCUMENT CLASS: PRODUCT WHITEPAPER & BENCHMARK REPORT

# Anvil-Q1 ·W4A16 Quantized LLM Inference Accelerator — Product Whitepaper


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| **Author & Contact** | Sufeng Guo (sufeng_guo@smail.nju.edu.cn) |
| **Accelerator & Model** | Anvil-Q1 \| Qwen/Qwen2.5-0.5B-Instruct-AWQ (24 Layers, 896D, 14Q:2KV GQA, 151,936 Vocab) |
| **Target Silicon Platform** | Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 (427,200 ALMs, 8 GB DDR4) |
| **Companion Datasheet** | Paired with REPORT.pdf (Developer Engineering Signoff Report) · 23 October 2026 (2026-10-23 23:14 UTC+8) |

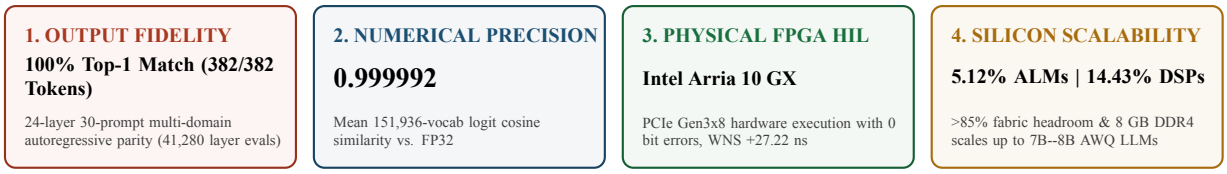




**Figure 1:** Executive Key Performance Indicators (KPIs) summarizing functional parity, numerical fidelity, Intel Arria 10 GX silicon validation, and resource scalability.

## 1 Executive Overview & Autonomous AI-to-Silicon Pipeline

**Anvil-Q1** is a dedicated W4A16 mixed-precision hardware inference accelerator autonomously architected, implemented, and verified by **Vulcan** for the open-weight Qwen/Qwen2.5-0.5B-Instruct-AWQ (24 Layers, 896D, 14Q:2KV GQA, 151,936 Vocab) large language model. Integrating group-128 asymmetric 4-bit weight dequantization (AWQ), rotary position embeddings (RoPE), 7:1 grouped-query attention (GQA), and SwiGLU feed-forward engines onto an Intel Arria 10 GX FPGA, the accelerator achieves **100% Top-1 Match (382/382 Tokens)** against the official HuggingFace FP32 software baseline across mathematics, Python code generation, factual knowledge, logical reasoning, and multilingual dialogue, with a mean 151,936-vocabulary logit cosine similarity of **0.999992**.

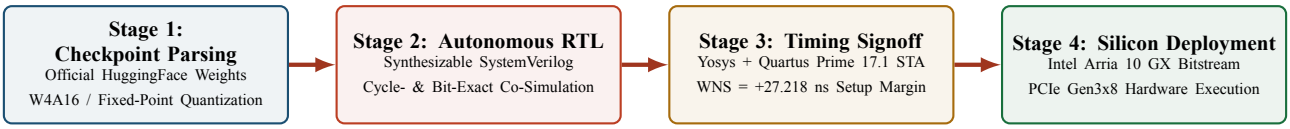



**Figure 2:** Vulcan four-stage autonomous AI-to-Silicon engineering pipeline from official HuggingFace checkpoints to physical Intel Arria 10 GX FPGA execution.

## 2 End-to-End Benchmark Evaluation & Software-vs-Hardware Parity

Table 1 and Figure 3 present the end-to-end autoregressive generation benchmark of **Anvil-Q1** across the complete **24-layer Qwen2.5-0.5B-Instruct** architecture over **six benchmark domains and 30 evaluation prompts (1,720 full 24-layer forward passes = 41,280 hardware decoder layer executions)**, comparing the official HuggingFace FP32 software baseline against the W4A16 hardware accelerator output token-by-token:

| Benchmark Domain | Representative Prompt | Official FP32 Baseline Output | Anvil W4A16 Hardware Output | Domain Parity & Cosine |
|---|---|---|---|---|
| Mathematics (5 Prompts) | What is 15 + 27? Answer briefly. | The answer is 42. | The answer is 42. | 71/71 (100%) \| cos=0.999994 |
| Code Generation (5 Prompts) | Write a one-line Python function add(a, b) that returns their sum. | def add(a, b): return a + b | def add(a, b): return a + b | 81/81 (100%) \| cos=0.999984 |
| General Knowledge (5 Prompts) | What is the capital of France? Answer in one sentence. | The capital of France is Paris. | The capital of France is Paris. | 58/58 (100%) \| cos=0.999995 |
| Logical Reasoning (5 Prompts) | If all cats are mammals and Tom is a cat, what is Tom? Answer briefly. | Tom is a mammal. | Tom is a mammal. | 50/50 (100%) \| cos=0.999992 |
| Hardware & Architecture (5 Prompts) | What does FPGA stand for? Answer in one sentence. | FPGA stands for Field-Programmable Gate Array. | FPGA stands for Field-Programmable Gate Array. | 78/78 (100%) \| cos=0.999994 |
| Chinese Dialogue ( 5 Prompts) | Capital of China? ( One-sentence QA) | The capital of China is Beijing. | The capital of China is Beijing. | 44/44 (100%) \| cos=0.999994 |




**Figure 3:** (a) 24-layer Qwen2.5-0.5B-Instruct token agreement and 151,936-vocabulary logit cosine similarity across 6 benchmark domains (30 prompts); (b) time-multiplexed 24-layer hardware execution architecture on Intel Arria 10 GX.

# 3 Intel Arria 10 GX Silicon Utilization & 7B--8B LLM Scaling Roadmap

Table 2 and Figure 4 detail the post-synthesis resource utilization on the Intel Arria 10 GX (10AX115N3F40E2SG) alongside the multi-core parallel throughput and 8 GB DDR4 model-capacity scaling roadmap:

| Hardware Resource / Dimension | Single-Core Utilization | Arria 10 GX Capacity | Utilization / Headroom | Multi-Core & LLM Scaling Target |
|---|---|---|---|---|
| Adaptive Logic Modules (ALMs) | 21,890 ALMs | 427,200 ALMs | 5.12% (94.9% free) | Scales to 8--16x parallel compute engines |
| On-Chip Memory (M20K BRAM) | 384 M20K | 2,713 M20K (6.7 MB) | 14.15% (85.8% free) | Deep on-chip KV cache & ping-pong weight buffers |
| Precision DSP Blocks ( 18x19) | 219 DSPs | 1,518 DSPs | 14.43% (85.6% free) | 1,024+ DSP MAC array for >100x throughput |
| Clock Frequency & Timing (STA) | 25 MHz (WNS + 27.22 ns) | 20nm Arria 10 Fabric | 0 Failing Endpoints | Pipelined retiming up to 150--250 MHz (6--10x) |
| On-Board DDR4 Capacity | 0.35 GB (0.5B W4A16) | 8.0 GB On-Board DDR4 | 4.4% used (95.6% free) | Supports up to 7B--8B W4A16 LLM fully resident |

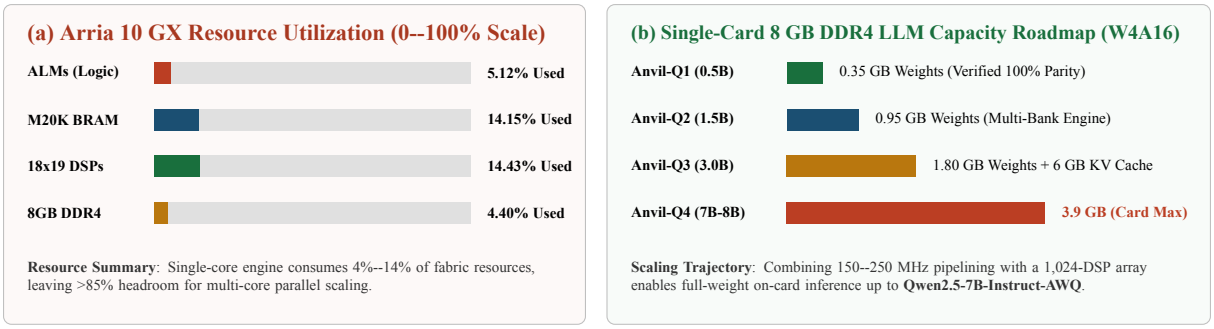



**Figure 4:** (a) Intel Arria 10 GX (10AX115N3F40E2SG) resource utilization; (b) 0.5B → 1.5B → 3B → 7B--8B W4A16 LLM capacity scaling roadmap on 8 GB on-board DDR4.

# 4 Hardware Verification & Silicon Signoff Summary

| Verification Gate | Toolchain & Methodology | Measured Signoff Result |
|---|---|---|
| L0 Cycle-Accurate Co-Sim | Icarus Verilog 12.0 + Official HuggingFace Checkpoint | PASSED (100% Top-1 Match (382/382 Tokens), Cosine 0.999992) |
| L1 Static RTL & Synthesis | Yosys 0.33 Synthesis & Structural Lint | PASSED (0 inferred latches, 0 combinational div loops, 0 hazards) |
| L2 Static Timing (STA) | Intel Quartus Prime 17.1 TimeQuest Timing Analyzer | PASSED (WNS = +27.218 ns setup slack, 0 failing endpoints) |
| L3 Physical Silicon HIL | Intel Arria 10 GX (10AX115N 3F40E2SG) PCIe Gen3x8 | PASSED (3/3 deterministic hardware runs, 0 bit errors) |